Projectile Tracking Chronograph (PTC)

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Chapter 2 - Project Description

2.1 Background and Motivation

The ability to accurately measure projectile velocity has long been fundamental to the study of ballistics and the advancement of shooting sports. Early methods relied on manual observation and empirical estimation, which were often inconsistent and dependent on environmental conditions. The development of mechanical ballistic chronographs in the late nineteenth century introduced a scientific approach by measuring velocity from distance over time, laying the groundwork for modern ballistic instrumentation.

In contemporary applications, ballistic chronographs have become essential tools for competitive shooters, hunters, and militaries as they provide quantitative data about projectile performance. Even small variations in muzzle velocity can significantly impact a projectile's trajectory and bullet drop, specifically at long ranges. Competitive shooters depend on accurate velocity measurements to refine ballistics and ensure the target is consistently hit. Other measurements of extreme spread and standard deviation are of interest, where lower variability in these metrics directly correlates with tighter shot grouping and improved accuracy.

Modern commercial chronographs further enhance usability by performing real-time calculations of muzzle energy, allowing users to simply gather projectile data rather than manual processing. However, many advanced chronograph systems rely on technology that can be expensive for hobbyists. Additionally, the commercially available systems focus primarily on velocity measurement and provide limited insight about the projectile's trajectory without the use of specialized or high-cost radar technology.

The PTC project is motivated by the need for an accessible and transparent ballistic measurement system that goes beyond the basic velocity calculation. By leveraging optical sensing, precise timing, and embedded computation, PTC aims to provide accurate projectile velocity measurements while enabling trajectory estimation through angular and spatial analysis. Beyond its technical objectives, PTC serves as an interdisciplinary senior design experience; this effort brings together optical engineering expertise focused on laser optics and beam deflection, electrical engineering contributions centered on analog sensing and signal processing, and computer engineering responsibilities of embedded firmware, timing analysis, and user interface implementation. Through collaboration, the team aims to develop a ballistic measurement system while fostering teamwork and problem-solving skills essential for professional engineering practice.

2.2 Goals and Objectives

The PTC project is designed to extend traditional ballistic chronograph functionality by combining optical sensing, embedded computation, and user-friendly data presentation into a single system. The goals define the overall vision of the system and its intended capabilities, while the objectives describe measurable outcomes that guide

implementation. The stretch goals are identified to encourage innovation beyond the minimum functional scope of the project.

2.2.1 Goals

Basic Goals:

- Develop an optical-based system capable of detecting a projectile as it passes through a defined measurement region.
- Accurately measure projectile velocity using precise timing and optical sensing techniques.
- Ensure the system operates reliably in indoor environments without dependence on ambient lighting conditions.
- Provide calculated results, including projectile velocity, to the user within 10 seconds of projectile detection.
- Present projectile-related information in an intuitive manner suitable for novice users.

Advanced Goals:

- Integrate environmental sensing (such as temperature and humidity) to support improved ballistic analysis.
- Extend the system's measurement capabilities beyond velocity to include trajectory-related information, such as the direction of projectile travel.
- Display processed data on an integrated screen, enabling standalone operation without the need for external devices.

Stretch Goals:

- Reconstruct and visualize a three-dimensional representation of the projectile's trajectory.
- Determine whether a projectile's trajectory would intersect a target with a one-meter diameter at a distance of ten meters.
- Incorporate atmospheric effects such as wind, temperature, humidity, and pressure into trajectory and offset calculations based on user-input target distance.
- Support a wide range of projectile velocities, from as low as 5 m/s to as high as 100 m/s.
- Determine and report the compass direction of projectile travel using onboard orientation and navigation sensors.

2.2.2 Objectives

Basic Objectives:

- Implement an optical detection system that does not require precise alignment by the end user.
- Capture projectile event data using optical sensors, cameras, or laser-based detection methods.
- Present results to the user via an integrated display with minimal user interaction required.
- Utilize a microcontroller such as a Raspberry Pi-based platform to coordinate sensing, timing, computation, and user interface tasks.

Advanced Objectives:

- Develop a calibration procedure that is intuitive, such as using known reference objects with the camera's field of view.
- Synchronize optical detection events with strobe lighting and camera/image acquisition to capture multiple images of the projectile in flight.
- Implement image processing and mathematical analysis to extract spatial information from captured images.

Stretch Objectives:

- Integrate a magnetometer to determine the compass orientation of the device and infer the compass direction of the projectile motion.
- Use an accelerometer to determine the vertical orientation of the system and calculate the projection elevation angle.
- Combine orientation, environmental, and optical data to improve trajectory estimation accuracy.

2.2.3 Projectile Tracking Chronograph: Features/Functionalities

The PTC is designed to provide a comprehensive set of features that extend beyond the traditional ballistic chronograph by incorporating various subsystems such as optical sensing, embedded computation, and spatial analysis. PTC implements an optical light detection system (light gate) that identifies a projectile as it enters the sensing region. By interrupting a modulated beam from a light source, the system generates high-speed timing events that are captured using hardware interrupts on the microcontroller. These events can be used to enable accurate time-of-flight measurements across a known distance, allowing for precise velocity calculations. The use of a modulated carrier frequency along with photodetector(s) that have a sufficient bandwidth will allow the system to operate in indoor and ambient lighting conditions. Additionally, PTC uses an optical chamber with synchronized strobe light illumination and camera imaging to capture multiple instances of the projectile as it is moving through the chamber. Once the initial velocity is obtained from the light gate, the microcontroller activates the strobe flashes and camera exposures. This in turn produces a single image containing multiple shadows of the projectile, each corresponding to a known interval of time. Image processing algorithms can then be executed on a Raspberry Pi to extract the spatial position data from the shadows of these images, allowing for the direction of the projectile and its angular trajectory to be approximated.

This system also supports embedded ballistic and trajectory computation, combining the previous timing data and spatial measurements to calculate the projectile velocity, angular deviation, and estimated trajectory path through the measurement region. In our advanced and stretch goals, we would like to further enhance these calculations by incorporating other environmental data from other types of sensors. Integrating information from temperature, humidity, and pressure sensors can be used to further refine the ballistic calculations to increase the overall accuracy of the system by accounting for atmospheric effects. Furthermore, combining other sensors such as an accelerometer and magnetometer would allow the system to determine its own orientation. This capability would support other advanced features, like determining the compass heading of the projectile's travel path.

To enhance the usability of PTC, a user interface consisting of simple input controls and a display has been incorporated into the system. Processed ballistic results are to be presented to the user within seconds of the detection of the projectile. This interface is designed to be user-friendly while still providing meaningful data to the user. Collectively, these features make the PTC a powerful tool for projectile measurement and analysis.

2.3 Current Technologies and Relevant Projects

2.3.1 The First Patented Ballistic Chronograph

Early projectile velocity measurement can be traced back to the late nineteenth century through purely mechanical systems such as the Le Boulenge ballistic chronograph, which was patented in 1874. This device relied on electromagnetic release mechanisms to drop two metal rods at different times as a projectile passed through two spaced wire screens. Knowing the acceleration due to gravity and the fixed spacing between the screens allowed the projectile's velocity and flight time to be calculated. Although entirely mechanical, this approach established the fundamental concept of velocity measurement through precise timing, a principle that remains central to modern electronic and optical chronographs [1].

2.3.2 Optical Chronograph

These devices operate by detecting the passage of a projectile across two optical sensors spaced at an identified distance. As the projectile passes over each sensor, the light received through the sensor is interrupted, generating electrical signals that are time-stamped to compute the projectile’s velocity [2].

While optical chronographs are relatively affordable and straightforward to operate, they are sensitive to environmental conditions. Measurement errors are introduced when there are changes in sunlight intensity, shadows, or improper projectile positioning. These commercial chronographs only display the velocity data. PTC builds upon the fundamental principle of measurement used by optical chronographs, but seeks to mitigate environmental sensitivity by designing laser-based detection and precise electronic timing, enabling improved consistency and expanded trajectory analysis beyond simple velocity measurement.

2.3.3 Magnetic Chronographs

Magnetic chronographs measure projectile velocity by detecting disturbances in a magnetic field caused by a projectile passing near a sensor array. These systems are typically mounted directly to the firearm barrel and do not rely on external lighting conditions [2].

Magnetic chronographs introduce certain limitations despite their reliability. The attachment of the device to the barrel can slightly alter the harmonics of the barrel, affecting where the projectile impacts. Magnetic chronographs are not suited for spatial position movement since their purpose is to measure velocity close to the muzzle. In

contrast, PTC is designed as a non-contact measurement system, allowing projectile velocity and angular trajectory to be determined without physical interaction with the firing mechanism, preserving ballistic integrity.

2.3.4 Doppler Radar Chronographs

Doppler radar chronographs represent the most advanced of the commercially available projectile measurement systems. These devices transmit radar signals that reflect off the projectile during flight, allowing the system to continuously track velocity based on Doppler frequency shifts [2].

While radar-based chronographs offer excellent accuracy and ease of use, they are costly and rely on complex signal processing hardware, limiting user access to customization of raw data. PTC aims to explore an alternative approach by leveraging optical scanning techniques and real-time signal processing to achieve comparable spatial resolution at a lower cost. By using laser interruption timing and angular position mapping, PTC seeks to bridge the gap between basic optical chronographs and high–end radar systems.

2.4 Required Specifications

2.4.1 System Specifications

General System Specifications:

Parameter	**Description**	**Specification**
Minimum detectable projectile size	Minimum diameter of a spherical projectile that can be detected and characterized	< 15 mm
Minimum speed accuracy	The minimum accuracy of the speed detected of the projectile	< 5%
Maximum system response time	The maximum time from a projectile traveling through the system until information is displayed to the user	< 5 s
Maximum detectable projectile size	Maximum diameter of a spherical projectile that can be detected and characterized	> 50 mm
Minimum detectable speed	The minimum speed of a projectile that can be	< 10 m/s

	detected and characterized	
Maximum detectable speed	The maximum speed of a projectile that can be detected and characterized	> 50 m/s
Minimum angular resolution	The minimum accuracy of the angle detected of the path of the projectile	$< \frac{1}{10}$ radian
Maximum power requirements	The maximum power the system will draw on average over the course of 10 seconds	< 1000 W
Sensing area	The area of the system that will be able to sense projectiles through it	>10 cm x 10 cm

Table 2.4.1.1 Specification of System Components

2.4.2 Component Specifications

Specific Component Specifications:

Component(s)	**Description**	**Specification**
Camera(s)	The sensor area, the quantum efficiency of detecting the photons in the wavelength of the strobe, and the resolution of the sensor	Large enough sensor size and quantum efficiency to be able to image the box at the given wavelength, on-time, and brightness of the strobe lights. Large enough resolution to image 15 mm objects
Photodetector(s)	The responsivity of the photodiode in the wavelength of the emitting laser or LED wavelength, the noise floor of the photodiode, and the bandwidth of the photodiode	> 100 kHz bandwidth Sufficient responsivity and signal to noise ratio to allow detection of the projectile
Gate light source(s)	The light source for the initial light gate for detecting the	Sufficient brightness to provide a good

	projectile entering into the system. Either a laser or LED.	signal to the photodetector(s) Sufficient bandwidth for providing the carrier signal for the light gate
Environmental Sensor(s)	Accuracy of the humidity, temperature, and pressure sensors	< ±10hPa accuracy at 1000 hPa < ± 5% humidity at 30%-80% humidity < ± 5℃ at 10℃-50℃
Power Supply	Provides a regulated voltage and current to power the electronic devices	Sufficient power to provide a constant current of ~50 A
Strobe Light Mechanism	Capabilities of the subsystem that creates the strobe flashes	< 200 µs flash time < 100 µs turn on time < 100 µs turn off time > 1000 lumen output
Physical System Structure	Physical layout of the system	>10 cm width and height sensing area <2 m sensing area length <3 m overall height <2 m width < 3 m overall length
User Interface	User interface specifications	> 2 cm width and height of screen <1 m width and height of screen
Microcontrollers/Computers	Microcontroller and computer specifications	> 1MHz MCU clock speed Image processing capability for computer

Table 2.4.2.1 Specification of Project Components

2.5 Optical Hardware

The system will consist of a light gate (or sensor gates) and an area for strobe photography of the projectile (optical chamber). The initial light gate will get a projectile speed that will determine the timing of the strobe flashes. The camera(s) will take one photo that contains multiple shadows of the projectile at the points in time where the strobe is triggered.

The cameras, lenses, and strobe LEDs depend on each other. The LEDs must be bright enough to drown out the ambient light leaking into the box in order to provide a good signal to the cameras. The lens system must be designed to capture enough light to provide a sufficient signal to show the projectile on the cameras. The LEDs will need to flash for a short enough time to avoid too much blurring of the projectile in the chamber; however, they need to flash for long enough to provide sufficient light to activate the camera sensor enough to get an acceptable picture. A rough estimate of a camera that operates at 30 fps can comfortably film at typical bedroom levels of illumination, which is about 10 lumens/square foot. Given an approximate strobe area of 2 square feet, and about 20% illumination efficiency, that would be 100 lumens for the 1/30 s or 33 ms that a camera frame lasts. With a 100 microsecond strobe time, that's 330 times the brightness needed for an equivalent amount of light to hit the camera sensor during the 100 microsecond strobe, which gives at least 3000 lumens required from the strobe for a simple camera.

The initial light gate will be used to detect the projectile entering the system and its approximate speed. The photodiode must be of sufficient bandwidth to detect a fast-moving projectile (a paintball is about 17mm and travels at 90 m/s, which gives 180 microseconds of signal time) would require at least 5 kHz of bandwidth. A projectile that's ¼ inch long and traveling at 1000 feet/s would require a 50 kHz bandwidth sensor to detect the projectile. In order to filter ambient light, a carrier wave of sufficiently higher bandwidth would be required, so 100 kHz would be the minimum for detecting a bullet.

Below, the diagram [2.5.1] elaborates on the design of the sensor gates and optical chamber responsible for detecting a projectile and taking a picture of the projectile as it moves through the enclosure. By interrupting the line of sight of the light source to its corresponding photo-sensor, a projectile can be detected for initial calculations to be made. As the projectile moves through the optical chamber, multiple cameras can be used to make further calculations that estimate position and trajectory.

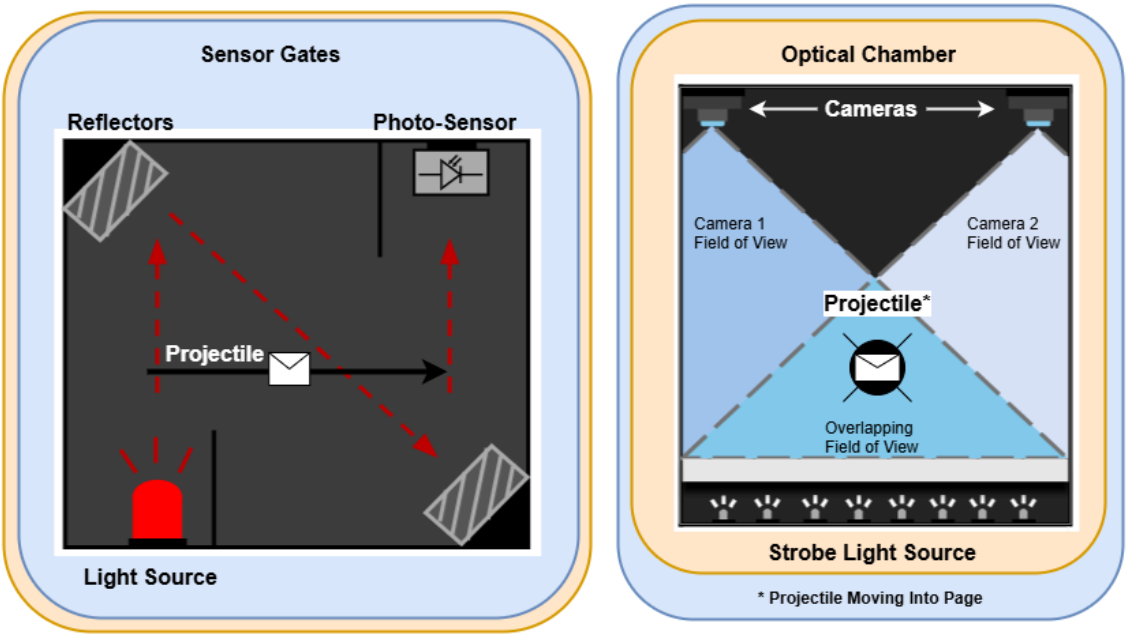


Figure 2.5.1 Diagrams elaborating on the sensor gates and optical chamber of the PTC system, illustrating the interaction between external projectile input and the optical system in place.

2.6 Electronics and Software Diagrams

The figure below [2.6.1] illustrates the necessary hardware and connections to facilitate our goals and objectives, as well as their current status at the time of this writing. Each block in the diagram is color-coded to indicate which team member will be primarily responsible for its research, design, and operation. The RP2040 MCU will be primarily responsible for handling sensor information and hardware-level interactions, whereas the Raspberry PI computer will be used for image processing of the projectile and to control the user interface. An optical sensor and chamber will be used to detect a projectile and take pictures, and these signals will be used both by the RP2040 and the Raspberry PI to make calculations.

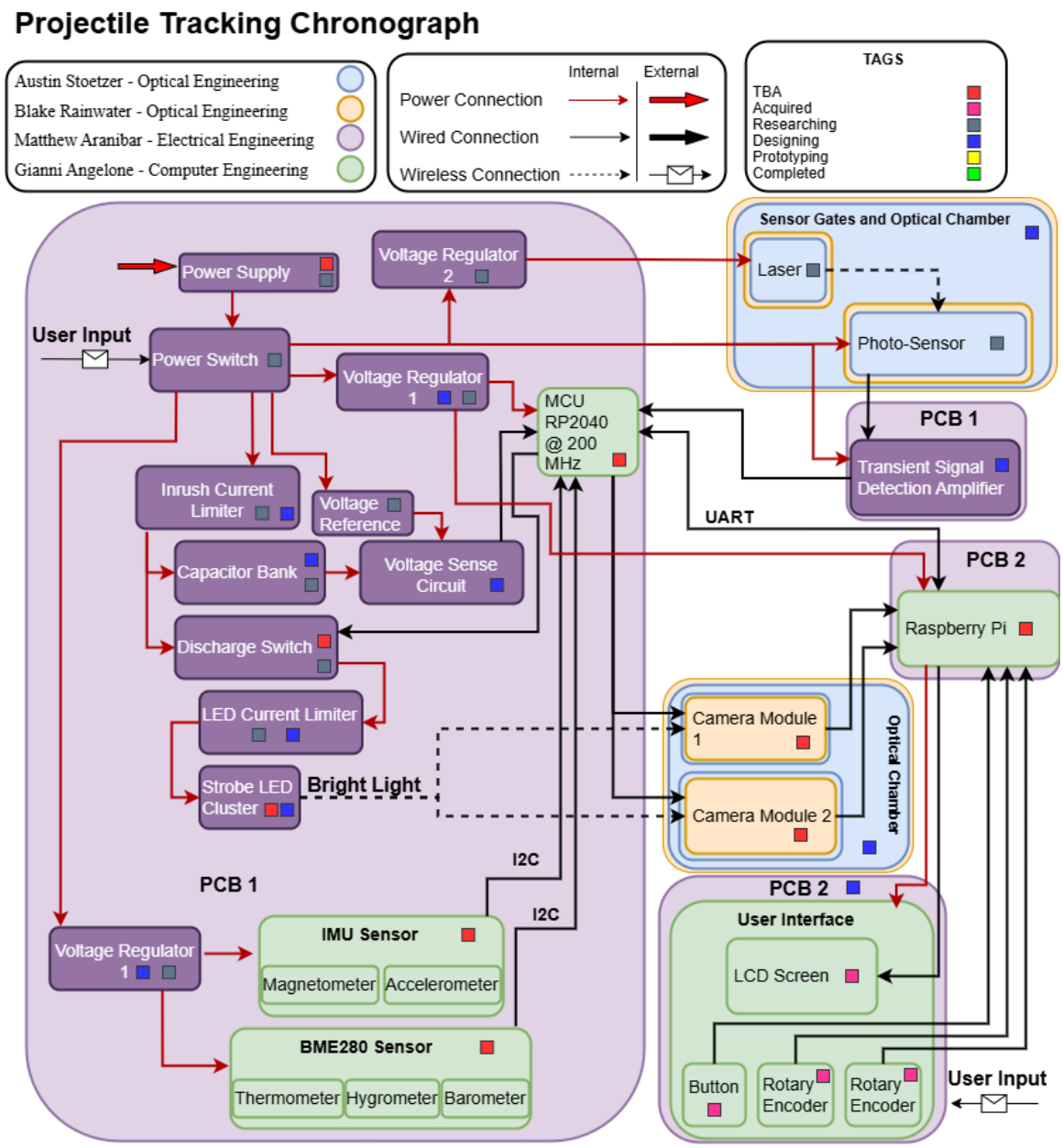


Figure 2.6.1 Hardware Diagram of the PTC system that illustrates component connections, status, and division of workload.

The figure below [2.6.2] presents the flowchart for the software portion of the PTC system, detailing the flow of data from initial projectile detection to final system output. The diagram outlines how optical interruptions at the gates trigger hardware interrupts that prompts the MCU to synchronize the cameras and LED strobes, enabling the capture of multiple images of the projectile as it moves through the system. These images are then processed using computer vision techniques to extract positional data, allowing for the velocity and trajectory to be calculated in the ballistic and trajectory engine. In parallel to that process, environmental sensor data are also being incorporated into the

overall computations. The resulting data from these subsystems is then communicated to the user interface, where the results are displayed.

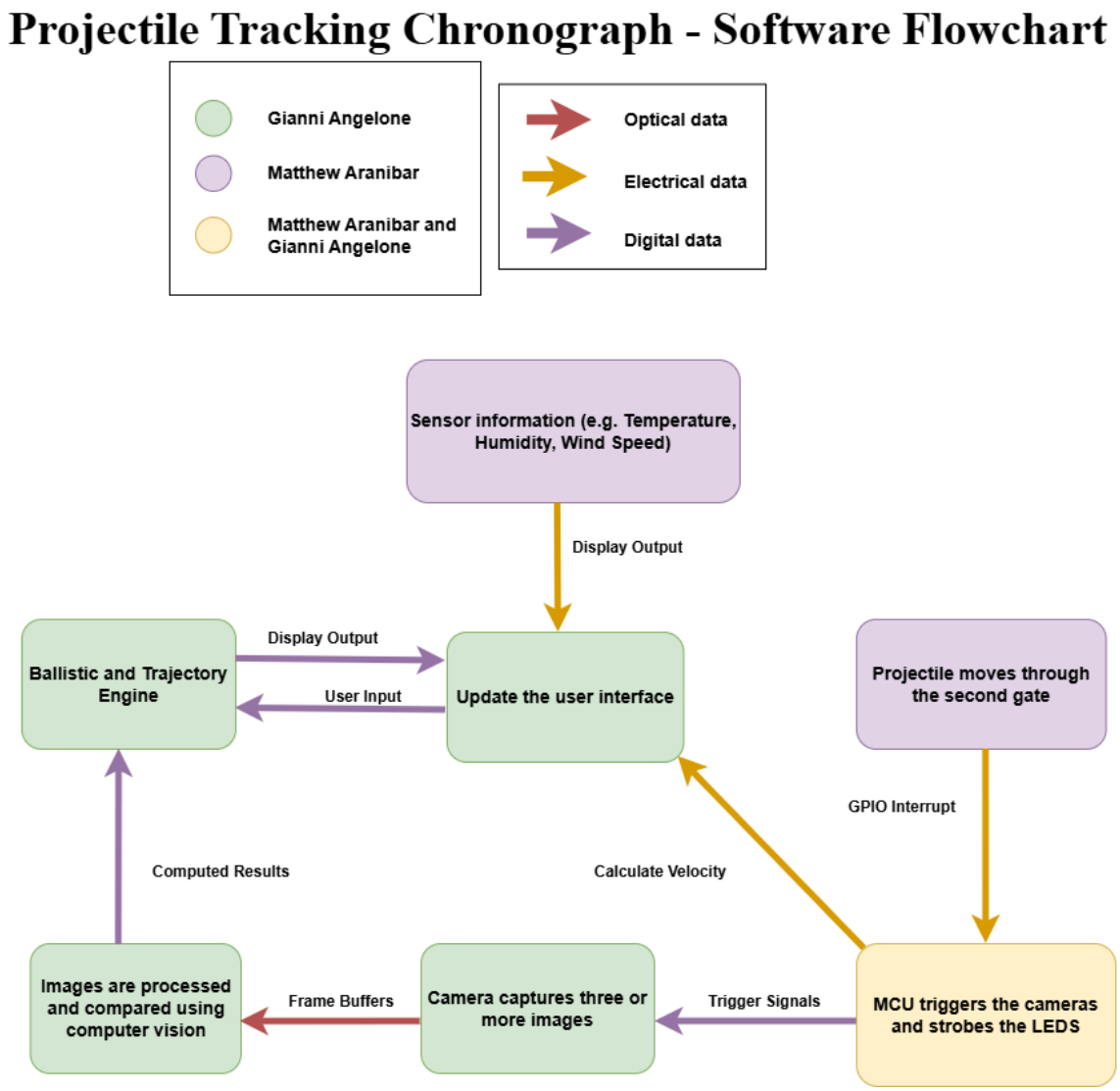


Figure 2.6.2 Software flowchart of the PTC system that illustrates the end-to-end data flow from optical event detection and hardware interfacing through synchronized strobe and cameras, image processing, and ballistic computation, outputting to the user via the user interface.

The next diagram [2.6.3] is a House of Quality showing the relationships (main body) between various customer and engineering requirements, as well as the correlations between those engineering specifications (roof). The relationships in the main body between customer and engineer have different weights depending on how closely one affects the other, with the strongest relationship having a weight of 7 and the weakest having a weight of 1. Each requirement also has a polarity, denoted by an up or down arrow, that displays the desirability said requirement. For example, “Affordability” would have a positive polarity because a cheaper product is a more desirable trait. In the roof of

the House, the correlations between engineering tradeoffs can be seen. A positive correlation means that improving one engineering requirement also benefits another, while a negative correlation means that improving one engineering requirement comes at the detriment of another. A neutral correlation means there is no tradeoff between two requirements if one is improved.

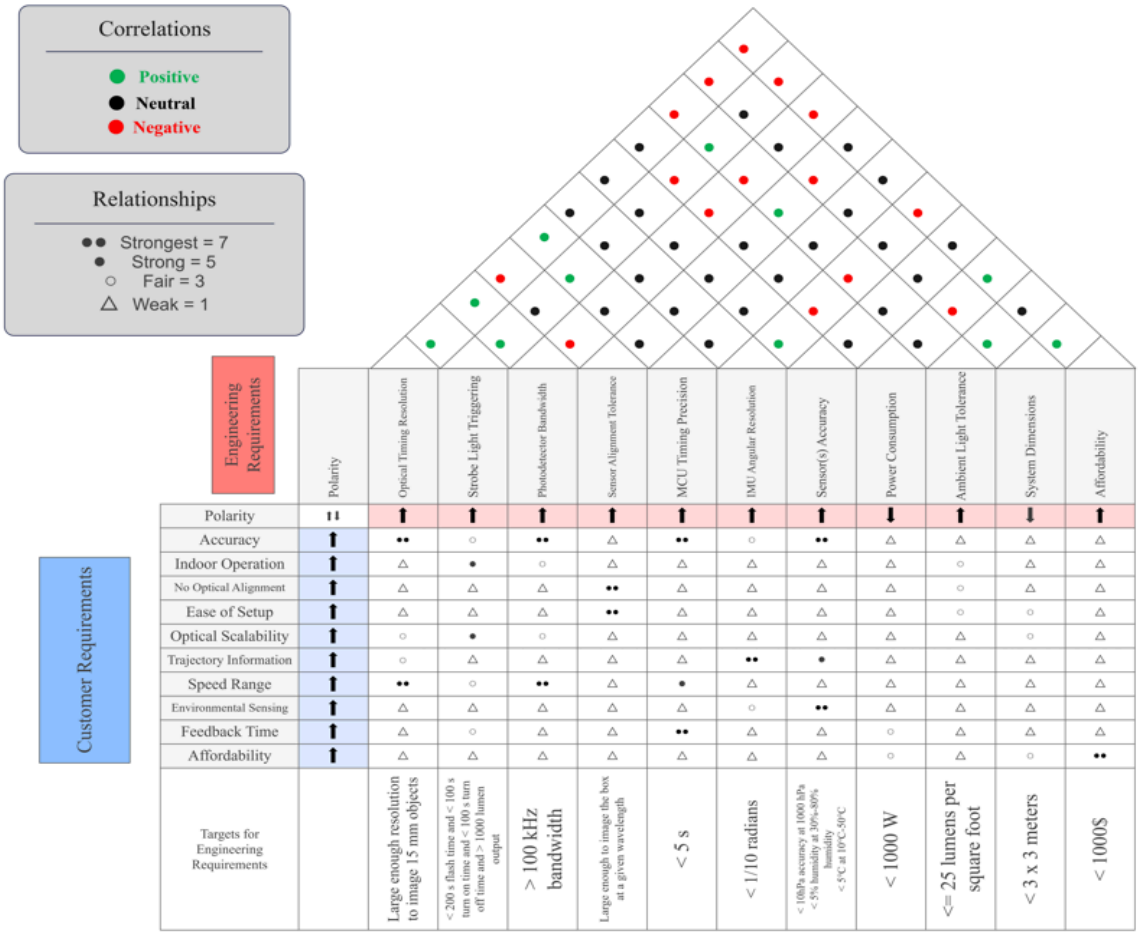


Figure 2.6.3 House of Quality diagram that shows the tradeoffs between the engineering and customer requirements of the project. The customer requirements (left-most column) depict the features that are more important to the user, while the engineering requirements (top row) show the features that are considered for various engineering aspects of the project.

Chapter 10 - Administrative Content

10.1 Budget

The PTC project is planned with a total expenditure limit of $1000, as established by the project team to ensure the system can be developed within reasonable cost constraints while still meeting technical objectives. This budget is intended to support the optical, electrical, and computing components that are required for the implementation of the optical projectile measurement system.

10.2 Bill of Materials (BOM)

The BOM provides a comprehensive list of the components that are necessary to design and build the PTC. Maintaining the BOM throughout the project will aid in cost awareness and serve as a centralized reference for the various hardware needs of the project. Below is a table [10.2.1] containing the components and estimated costs of each part.

Components	Estimated Cost
Camera (x2)	30-50$
Photodetector(s)	~10$
Gate light source(s)	~120$
Power Supply	50-124$
Strobe LED Cluster	~13$
Physical System Structure	25-50$
Raspberry Pi	100-140$
LCD Screen	40-50$
Button	0.13$
Rotary Encoder (x2)	~1.50$
IMU Sensor	12.34$
BME280 Sensor	12-18$
MCU RP2040	~3$
Estimated Maximum Total Cost: 580$	

Table 10.2.1 Bill of Materials

10.3 Project Milestones

This senior design project involves multiple interconnected technical and documentation tasks, so setting clearly defined milestones with set deadlines is essential for maintaining consistent progress and organization. Clearly scheduled milestones break the project into manageable phases, helping our team to prioritize either individual or group tasks at any given time. By committing to specific dates, our team is able to ensure that all progress made between each stage of the project is done so in a controlled and predictable manner, thereby minimizing any last-minute risks that may occur. Below is our project milestone

table [10.3.1] that has a description of each milestone, the start/anticipated end dates, and each milestone's dependencies.

Milestone #	Milestone Description	Start Date	Anticipated End Date	Dependencies
1	Finalize Project Concept and Scope	1/27/26	1/29/26	Advisor Approval
2	Divide and Conquer Document Finalization	1/25/26	1/29/26	Team role definition
3	Divide and Conquer Review Meeting	2/3/26	2/3/26	Completed D&C documentation
4	Committee Formation and Initial Meeting	1/26/26	1/30/26	Faculty availability
5	30-Page Draft Development	2/3/26	2/17/26	D&C revision and expansion
6	30-Page Draft Revision	2/17/26	2/20/26	Continued progress of project and following of guidelines
7	60-Page Draft Development	2/20/26	3/6/26	Continued progress of project and following of guidelines
8	60-Page Draft Revision	3/6/26	3/10/26	Continued progress of project and following of guidelines
9	90-Page Draft Development	3/10/26	3/24/26	Continued progress of project and following of guidelines
10	90-Page Draft Revision	3/24/26	3/27/26	Continued progress of

				project and following of guidelines
11	120-Page Final Report	3/27/26	4/10/26	Review of team parameters and follow document guidelines
12	Mini Demonstration Video	4/28/26	4/28/26	Hardware and software integration

Table 10.3.1 Project Milestones

10.4 Hardware Milestones

The hardware portion of this senior design project involves several design and testing steps, making explicitly defined hardware milestones essential for maintaining steady progress. Since there are many interdependent portions of this project, having set deadlines for milestones such as gathering components and prototyping will ensure that everyone in the group is able to make consistent advancements in their respective portions. Below is the hardware milestone table [10.4.1], with the start/anticipated end dates as well as a description of each milestone.

Hardware Milestone	Start Date	Anticipated Completion Date	Description
Component Selection	1/26/26	2/1/26	Making a BOM
Gather Optical Components	1/28/26	2/5/26	Ordering from a manufacturer
Gather Camera(s)	2/3/26	2/7/26	Ordering from a manufacturer
Gather Sensors	2/3/26	2/7/26	Ordering from a manufacturer
Find image processing software	1/28/26	2/3/26	Researching available options
Start Prototyping	2/3/26	2/15/26	All components are tested and work
Complete Project with Prototype Board	3/1/26	4/1/26	Assemble the components for a

			prototype
PCB Testing	4/5/26	4/10/26	Creating and ordering a customized PCB.
Test Prototype with PCB	4/10/26	4/14/26	Using the customized PCB for the design
Finalize Prototype	SD2	SD2	Finalize the design using a specialized PCB board

Table 10.4.1 Hardware Milestones

Appendicies

Appendix A - Citations

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Appendix B - Useful Tools

[1] EdrawMax Online
[2] Draw.io
[3] Microsoft PowerPoint
[4] Google Docs
[5] Digikey
[6] Google Search Engine
[7] Ultra Librarian
[8] Snap Magic Search